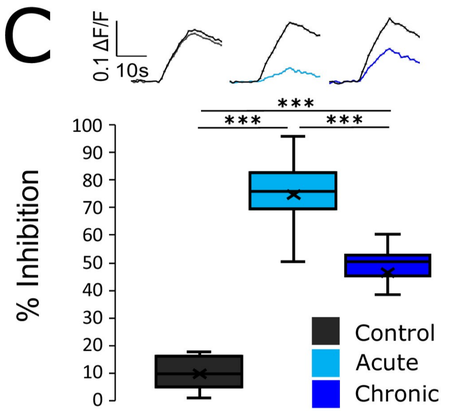
Version A · three teaching bullets with sub-bullets

Opioids reduce release through a presynaptic receptor

- **Opioid receptors** can reduce communication at a synapse by acting on the **presynaptic terminal**, the end of the axon that releases neurotransmitter.
  - In cultured neurons, electrical stimulation normally causes **synaptic vesicles** to fuse with the terminal membrane and release their contents.
  - **DAMGO** is a drug that activates **mu opioid receptors**. With DAMGO present, the same stimulation caused **less vesicle fusion**, so less transmitter was released.
- Release was measured with a **fluorescent protein attached to a vesicle protein**.
  - The fluid inside a vesicle is acidic, and in acid the protein glows only weakly.
  - When a vesicle fuses, the protein meets the fluid outside the cell and becomes **brighter**. A smaller rise in brightness means **fewer vesicles fused**.
- After **prolonged DAMGO treatment**, the neurons developed **tolerance**. DAMGO reduced release by about **75%** in untreated neurons, but by only about **50%** after chronic treatment.
  - The **dose–response curve shifted**, so more DAMGO was needed to produce the same reduction.
  - The **largest possible inhibition also decreased**, so even high doses could not block release as strongly.



Jullié et al. (2022), Fig. 1 (C). Presynaptic opioid responses after chronic treatment.

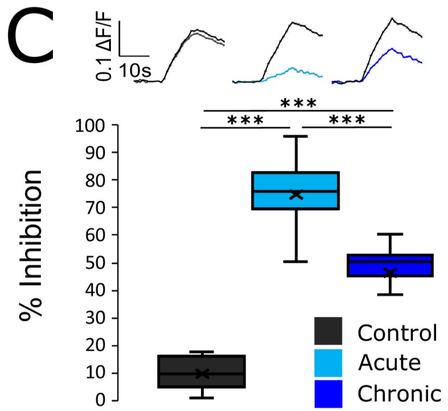
Version B · three teaching paragraphs

Opioids reduce release through a presynaptic receptor

Opioid receptors can reduce communication by acting on the **presynaptic terminal**, the end of the axon that releases neurotransmitter. In cultured neurons, electrical stimulation normally causes **synaptic vesicles** to fuse with the terminal membrane. **DAMGO**, a drug that activates **mu opioid receptors**, reduced this fusion. The same stimulation therefore produced **less transmitter release** when DAMGO was present, so the receiving neuron got a weaker chemical signal.

Release was measured with a **fluorescent protein attached to a vesicle protein**. The fluid inside a vesicle is acidic, and in acid the protein glows only weakly. When a vesicle fuses with the membrane, the protein meets the fluid outside the cell and becomes **brighter**. A smaller rise in brightness after stimulation means that **fewer vesicles fused**.

After **prolonged DAMGO treatment**, the neurons showed **tolerance**. In untreated neurons, DAMGO reduced the release signal by about **75%**. After chronic treatment, the same DAMGO application reduced it by only about **50%**. The **dose–response curve also shifted**, so more DAMGO was needed to produce the same reduction. The **largest possible inhibition decreased**, so even high doses could no longer block release as strongly as in untreated neurons. This tolerance depended on **endocytosis**, the process that pulls receptors from the cell surface into the cell.

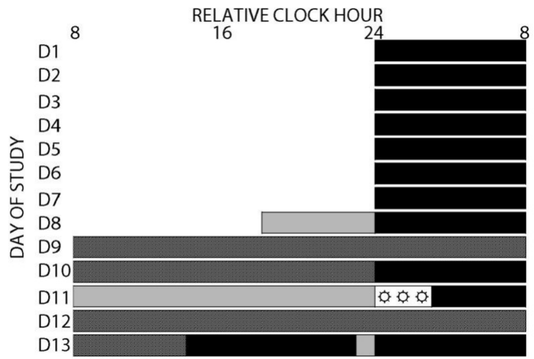


Jullié et al. (2022), Fig. 1 (C). Presynaptic opioid responses after chronic treatment.

Version A · three teaching bullets with sub-bullets

Evening caffeine can delay the human melatonin rhythm

- **Burke and colleagues** tested whether **evening caffeine** shifts the **circadian clock**, the internal timer that sets the body's roughly 24-hour daily rhythm.
  - Participants lived in a lab with **no clocks or daylight**. Melatonin timing was measured in a **40-hour constant routine**, awake in dim light, before and after treatment.
  - Each person received **caffeine or a placebo pill three hours before their usual bedtime**.
  - **Melatonin** is a hormone that rises in the evening and marks the start of the body's internal night.
- **Caffeine delayed the melatonin rhythm by about 40 minutes**, so the internal night began later.
  - **Bright evening light** also delayed the rhythm in a separate condition.
  - Caffeine had two effects: it made people **more alert that evening**, and it **moved their clock later**.
- Caffeine also **lengthened the daily rhythm** in **cultured human cells** carrying **clock reporters**, genes that make the cells glow in a daily rhythm.
  - The effect worked through **adenosine receptors**, which caffeine blocks, and **cyclic AMP**, a messenger molecule inside the cell.
  - Caffeine therefore acts on the **clock inside cells**, separately from keeping a person awake later.



Burke et al. (2015), Fig. 1, selected panels.

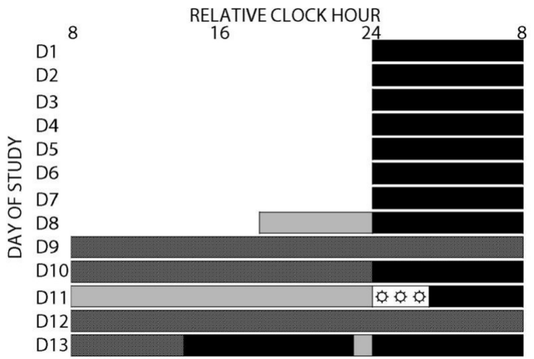
Version B · three teaching paragraphs

Evening caffeine can delay the human melatonin rhythm

Burke and colleagues tested whether **evening caffeine** shifts the **circadian clock**, the internal timer that sets the body's roughly 24-hour daily rhythm. Participants first kept a regular sleep schedule at home for about a week. They then lived in a laboratory with **no clocks or daylight** and stayed in **dim light** while awake. Each person received **caffeine or a placebo pill three hours before their usual bedtime**. The placebo was a matching pill filled with rice powder. The researchers tracked **melatonin**, a hormone that rises in the evening and marks the start of the body's internal night. Melatonin timing was measured in a **40-hour constant routine**, in which people stayed awake in dim light, once before and once after the treatment.

Caffeine delayed the melatonin rhythm by about 40 minutes, so the internal night began later. **Bright evening light** also delayed the rhythm in a separate condition. Caffeine therefore had two effects. It made people **more alert that evening**, and it **moved their clock later**.

Caffeine also changed the clock in **cultured human cells** carrying **clock reporters**, genes that make the cells glow in a daily rhythm. In these cells, caffeine **lengthened the molecular rhythms**. The effect worked through **adenosine receptors** and **cyclic AMP**, a messenger molecule inside the cell. Caffeine blocks adenosine receptors, which changes cyclic AMP signaling. Caffeine therefore acts on the **clock inside cells**, separately from keeping a person awake later.

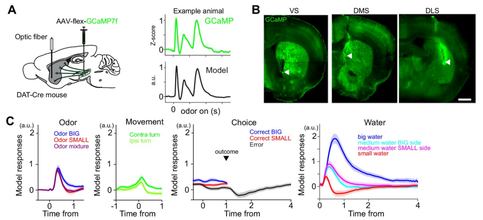


Burke et al. (2015), Fig. 1, selected panels.

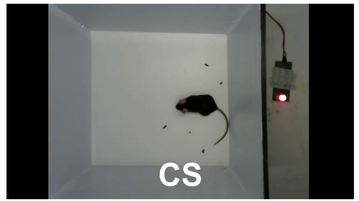
Version A · three teaching bullets with sub-bullets

Dopamine axons respond to changes in expected reward

- Researchers recorded **dopamine axons** in three parts of the mouse **striatum**: the **ventral**, **dorsomedial**, and **dorsolateral** striatum.
  - Dopamine neurons made a **calcium sensor** that glows when they are active, recorded through an **optic fiber** in the striatum.
  - In the task, a mouse smelled an **odor cue**, waited at least one second, and then chose one of two **water ports**. Each odor predicted a **big** or **small** amount of water.
- The axon signals changed at the **cue**, the **choice**, and the **outcome**, following changes in **expected reward**.
  - A **reward prediction error** is the difference between the reward an animal expects and the reward it gets.
  - A **temporal-difference model** calculates this error moment by moment, and its predictions matched the recorded signals.
- Dopamine axon activity carried information about **reward expectation**: a cue that **raises the expected reward** increases the signal.
  - A **big** water reward produced a large signal, and a **small** reward produced a **dip below baseline**.
  - A **medium** reward produced a larger signal where small rewards were expected than where big rewards were expected.
  - These error signals **update the animal's expectations** on later trials.



Tsutsui-Kimura et al. (2020), Fig. 2.



Feng et al. (2021), Video 3.

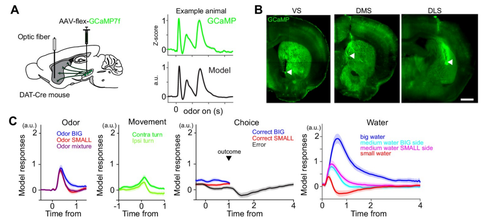
Version B · three teaching paragraphs

Dopamine axons respond to changes in expected reward

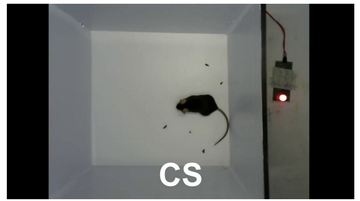
Researchers recorded **dopamine axons** in three parts of the mouse **striatum**: the **ventral**, **dorsomedial**, and **dorsolateral** striatum. A virus injected into the midbrain, where dopamine cell bodies sit, made these neurons produce a **calcium sensor** that glows when they are active. An **optic fiber** in the striatum collected the light from their axons. In the task, a mouse smelled an **odor cue**, waited at least one second, and then chose one of two **water ports**. Each odor predicted whether the correct port would give a **big** or **small** amount of water.

The axon signals changed at the **cue**, the **choice**, and the **outcome**, following changes in **expected reward**. A **reward prediction error** is the difference between the reward an animal expects and the reward it gets. A **temporal-difference model** calculates this error at each moment as the expectation is updated, and its predictions matched the recorded dopamine signals.

A cue that **raises the expected reward** increases the dopamine signal, so the big-water odor produced a larger signal than the small-water odor. When the water arrived, a **big** reward produced a large signal, and a **small** reward produced a **dip below baseline**. A **medium** reward produced a larger signal where small rewards were expected than where big rewards were expected. Dopamine axon activity therefore carries information about **reward expectation**. These error signals **update the animal's expectations** on later trials, which is how the signal supports learning.



Tsutsui-Kimura et al. (2020), Fig. 2.

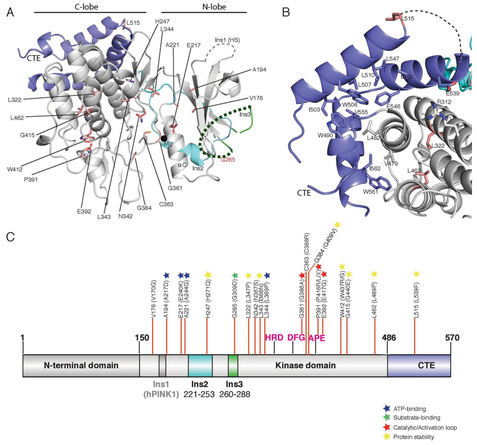


Feng et al. (2021), Video 3.

Version A · three teaching bullets with sub-bullets

Parkinsonian movement follows loss of nigrostriatal input

- **Parkinson's disease** causes **slowed movement**, **muscle rigidity**, and a **tremor at rest**, along with problems outside movement.
  - The disease destroys **dopamine neurons in the substantia nigra** of the midbrain.
  - Their axons reach the **striatum**, part of the **basal ganglia**, which help start and control movement. This is the **nigrostriatal pathway**.
  - Some inherited forms come from mutations in the enzyme **PINK1**, many in its **kinase domain**, which adds phosphate groups to other proteins.
- Dopamine acts on striatal neurons through **different receptor types**.
  - Through **D1 receptors**, dopamine excites neurons in a pathway that **helps start movement**.
  - Through **D2 receptors**, dopamine inhibits neurons in a pathway that **holds movement back**.
  - With less dopamine, the first pathway becomes less active and the second more active, so movements become **slow and hard to start**.
- **Levodopa** (L-DOPA) is a **precursor** that the remaining dopamine neurons convert into dopamine, which **improves movement symptoms**. It **does not replace the lost neurons**, which continue to die while symptoms are treated.



Kumar et al. (2017), Fig. 5.

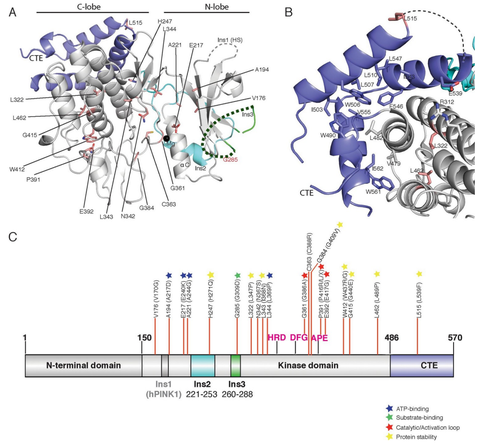
Version B · three teaching paragraphs

Parkinsonian movement follows loss of nigrostriatal input

Parkinson's disease causes **slowed movement**, **muscle rigidity**, and a **tremor at rest**, along with problems outside movement. The disease destroys **dopamine neurons in the substantia nigra**, a small region of the midbrain. These neurons send axons to the **striatum**, part of the **basal ganglia**, which help start and control movement. This connection is the **nigrostriatal pathway**. Some inherited forms of the disease are caused by mutations in the enzyme **PINK1**. Kumar and colleagues mapped these mutations onto the enzyme's structure, and many lie in its **kinase domain**, the part that adds phosphate groups to other proteins. Several sit next to the enzyme's **hydrophobic core**, the tightly packed center that holds the protein in its folded shape.

Dopamine acts on striatal neurons through **different receptor types**. Through **D1 receptors**, it excites neurons in a pathway that **helps start movement**. Through **D2 receptors**, it inhibits neurons in a pathway that **holds movement back**. When dopamine is lost, the first pathway becomes less active and the second becomes more active, so movements become **slow and hard to start**.

Levodopa (L-DOPA) is a **precursor**, a molecule that the remaining dopamine neurons convert into dopamine. Raising dopamine production **improves movement symptoms**. Levodopa **does not replace the lost neurons**, which continue to die while symptoms are treated.

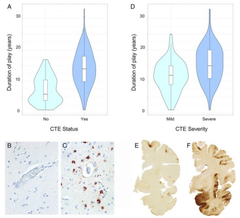


Kumar et al. (2017), Fig. 5.

Version A · three teaching bullets with sub-bullets

Repeated contact can occur without concussion

- A **concussion** is a **traumatic brain injury** that causes a sudden, short-term disturbance in brain function right after a hit.
  - **Repeated head impacts** include concussions and hits with **no obvious symptoms**, so a player can take **many hits to the head** without a concussion diagnosis.
- In ordinary football plays, players collide at the **line of scrimmage** and during **tackles**. Game footage records **how often** players collide.
  - **Helmet accelerometers** measure the **force** of each hit. Daneshvar and colleagues used them to estimate each player's **total head impacts** across a career.
  - Players with more estimated impacts had **higher odds of CTE**.
- **Cumulative exposure** depends on **how often** hits occur, **how hard** they are, and **how they are spread over time**.
  - A player's **concussion count** includes only hits that caused symptoms and were recognized, so researchers also measure **years of play**.
  - **CTE** (chronic traumatic encephalopathy) is a brain disease in which an abnormal form of the protein **tau** builds up in neurons. It is diagnosed by examining brain tissue after death.
  - In brain donors, those **with CTE** had played a median of about **14 years**, versus about **6 years** without CTE. Donors with **severe CTE** had also played longer than those with **mild CTE**.



Mez et al. (2019), Fig. 2.



Princeton–Yale football, 1903. Edison/Library of Congress (1903).

Play video

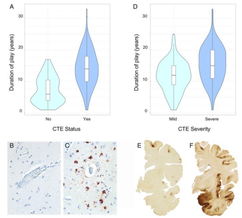
Version B · three teaching paragraphs

Repeated contact can occur without concussion

A **concussion** is a **traumatic brain injury** that causes a sudden, short-term disturbance in brain function right after a hit. **Repeated head impacts** include concussions and also hits that cause **no obvious symptoms**. A player can therefore receive **many hits to the head** without ever being diagnosed with a concussion.

In ordinary football plays, players collide at the **line of scrimmage** and during **tackles**. Game footage records **how often** players collide, while the **force** of each hit is measured with sensors such as **helmet accelerometers**. Daneshvar and colleagues used accelerometer data to estimate each player's **total head impacts** across a career. Players with more estimated impacts had **higher odds of CTE**.

Cumulative exposure depends on **how often** hits occur, **how hard** they are, and **how they are spread over time**. A player's **concussion count** includes only hits that caused symptoms and were recognized, so researchers also measure **years of play**. **CTE** (chronic traumatic encephalopathy) is a brain disease in which an abnormal form of the protein **tau** builds up in neurons. It is diagnosed by examining brain tissue after death, using an antibody that binds **phosphorylated tau**, the altered form that builds up in CTE. In brain donors, those **with CTE** had played a median of about **14 years**, compared with about **6 years** for those without CTE. Donors with **severe CTE** had also played longer than those with **mild CTE**, and each additional year of play was linked to higher odds of CTE.



Mez et al. (2019), Fig. 2.



Princeton–Yale football, 1903. Edison/Library of Congress (1903).

Play video